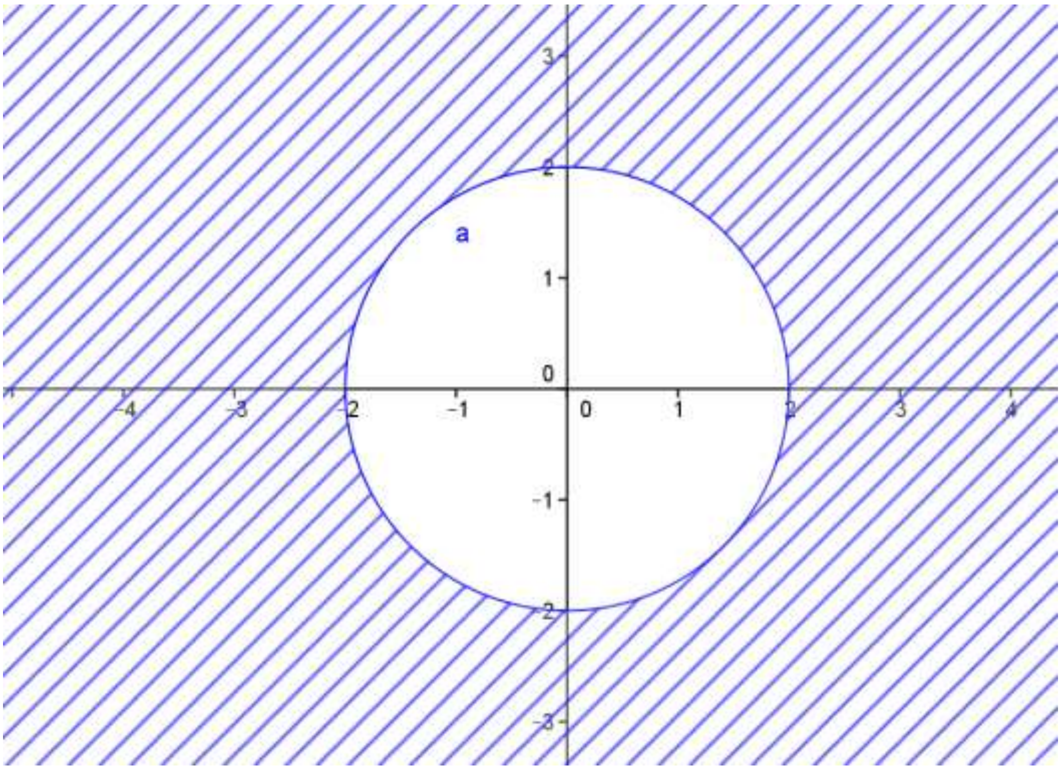
Resolución trabajo práctico. Funciones de dos variables independientes

RESOLUCIÓN DE LAS ACTIVIDADES

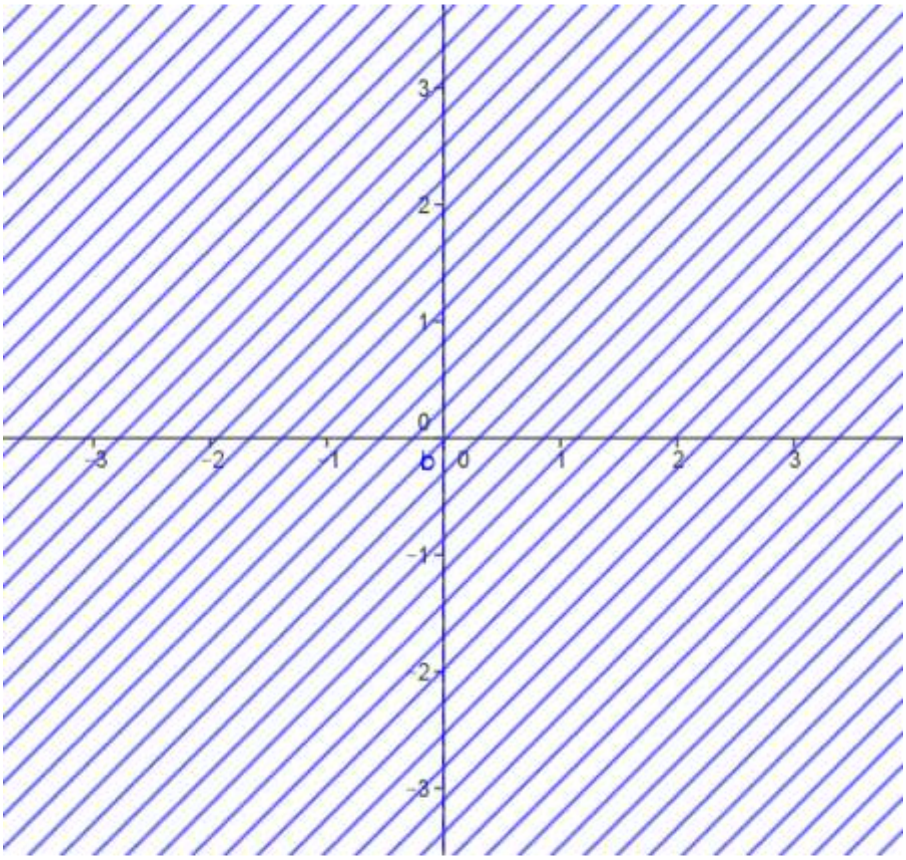
1.

a) $D = R^2$

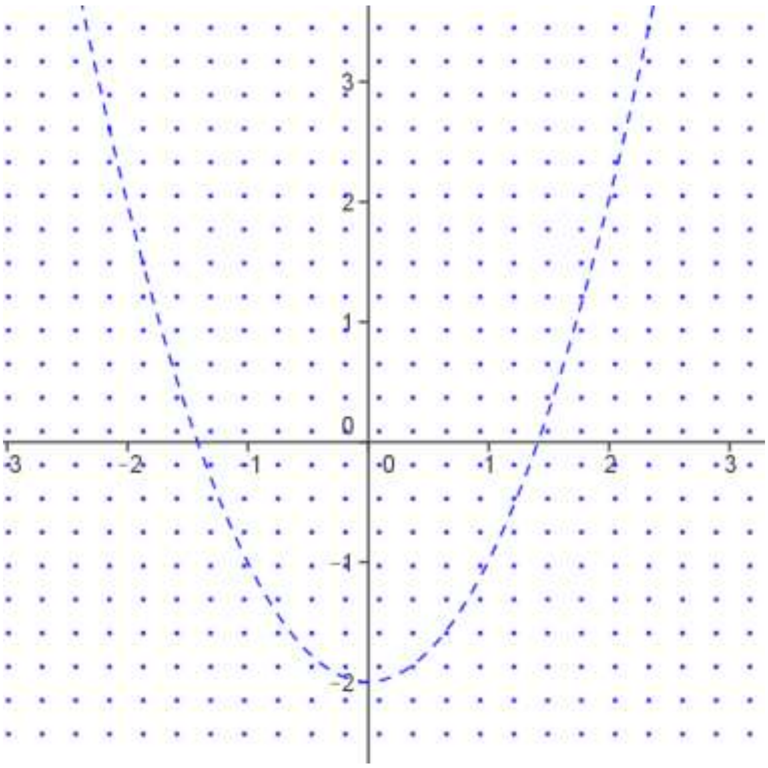
b) $D=\{(x,y) \in R^2 / x^2+ y^2 \geq 4\}$



c) $D=\{(x,y) \in R^2 / x \neq 0\}$



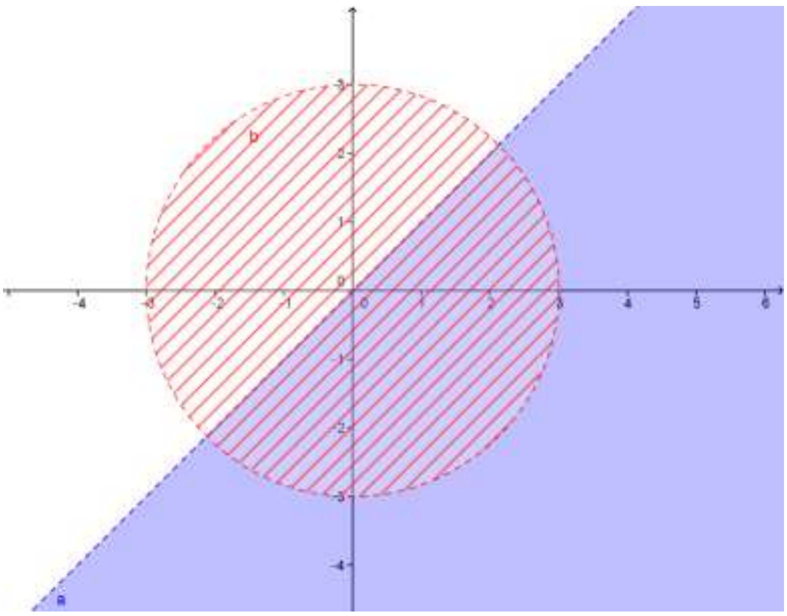
d) $D=\{(x,y) \in R^2 / x^2 - 2 \neq y\}$



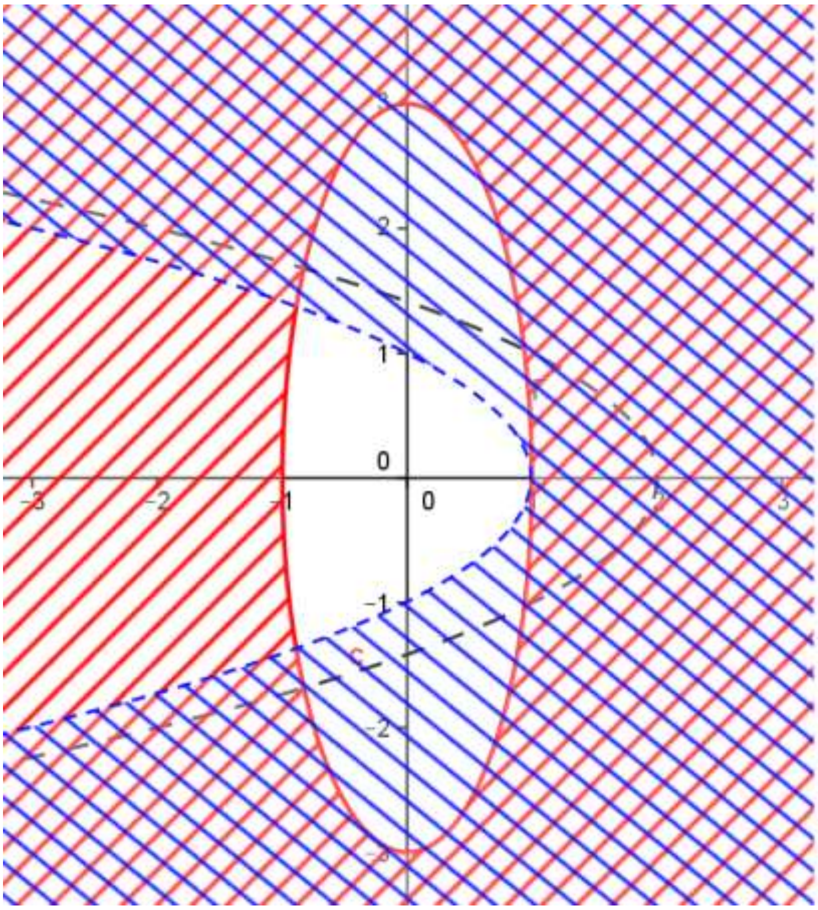
e) ---------------------

f) ---------------------

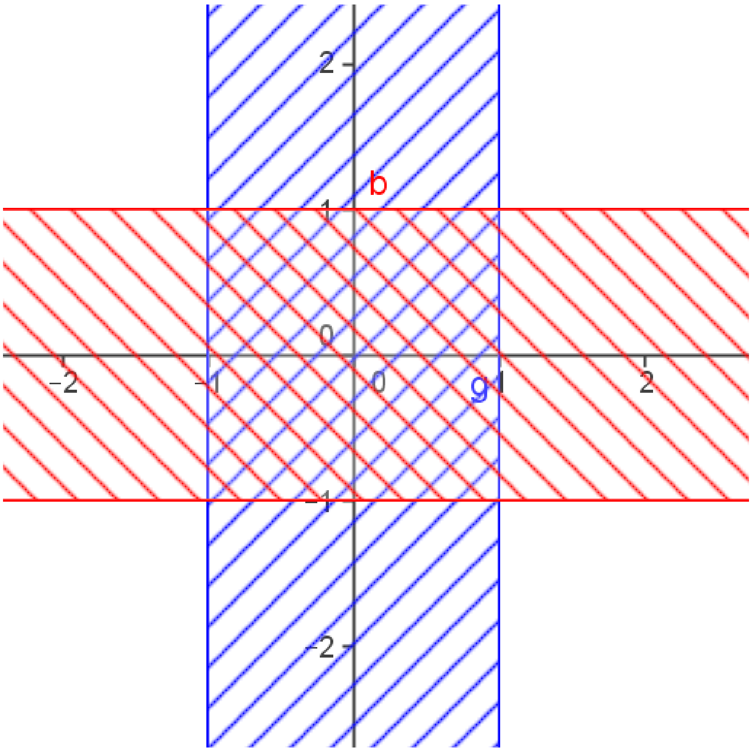
g) $D=\{(x,y) \in R^2 / x^2+ y^2 < 9 \wedge x < y\}$



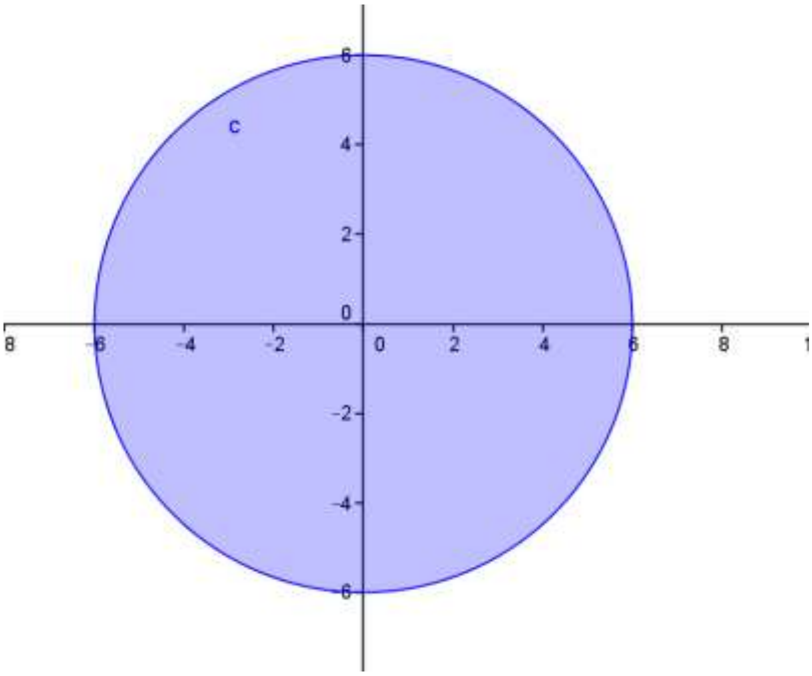
h) $D = \left\{(x, y) \in R^2 / \, x^2 \;+\frac{y^2}{9} \geq 1 \wedge \; y^2 + x > 1 \wedge y^2 + x \neq 2\right\}$



i) $D=\{(x,y) \in R^2 / x^2 \leq 1 \wedge y^2 \leq 1\}$



j) $D=\{(x,y) \in R^2 / x^2 + y^2 \leq 36\}$



2.

a) $F(x,y)=\frac{\ln(x+y-1)}{\sqrt{4-x^2-y^2}}$

b) $F(x,y)=\frac{\ln(y-x+1)}{\sqrt{4-x^2-y}}$

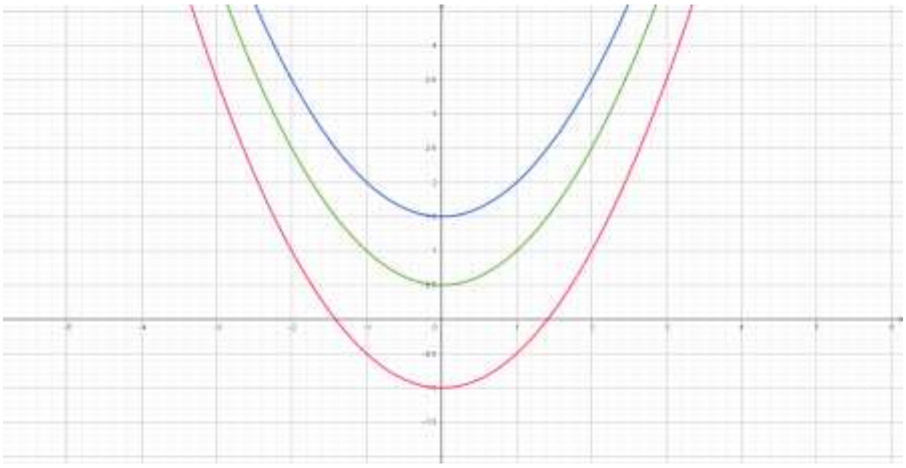
Curvas de nivel

3.

a) $z = 1 \;\rightarrow\; -x^2 + 2y = 1 \;\rightarrow y = \frac{1}{2} + \frac{1}{2}\,x^2$

$z = -2 \;\rightarrow\; -x^2 + 2y = -2 \;\rightarrow y = -1 + \dfrac{1}{2}\,x^2$

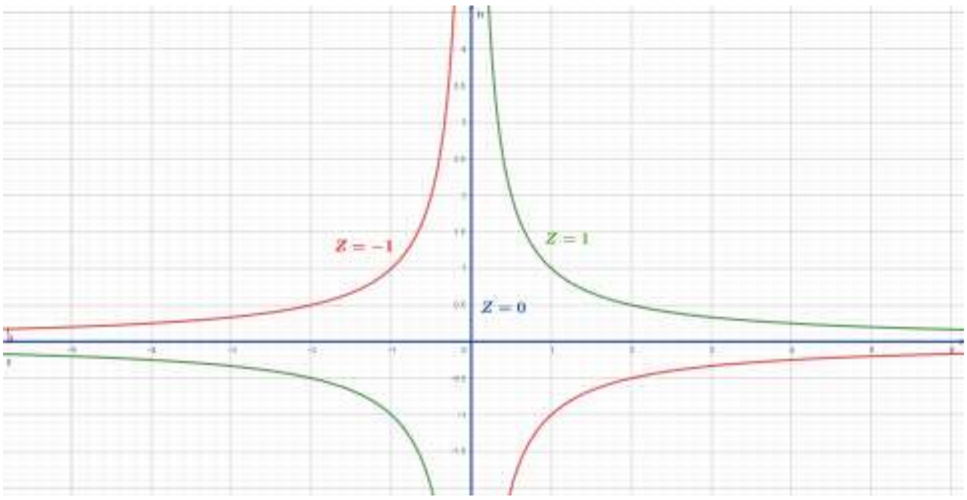
$z = 3 \;\rightarrow\; -x^2 + 2y = 3 \;\rightarrow y = \dfrac{3}{2} + \dfrac{1}{2}\,x^2$



b) $z = 1 \;\rightarrow\; x.y = 1 \;\rightarrow y = \frac{1}{x}$

$z = -1 \;\rightarrow\; x.y = -1 \;\rightarrow y = -\dfrac{1}{x}$

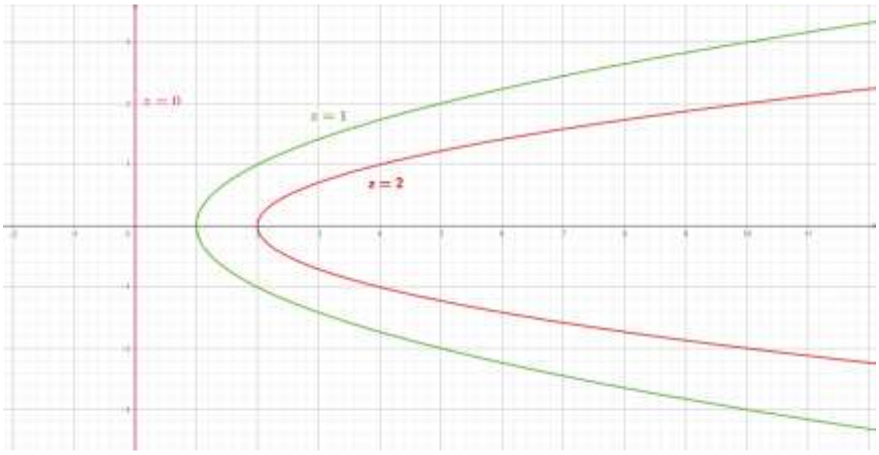
$z = 0 \;\rightarrow\; x.y = 0 \;\rightarrow x = 0 \;\vee\; y = 0$



c) $z = 1 \rightarrow \frac{x}{1+y^2} = 1 \rightarrow x = 1 + y^2$

$$z = 2 \rightarrow \frac{x}{1+y^2} = 2 \rightarrow x = 2 + 2y^2$$

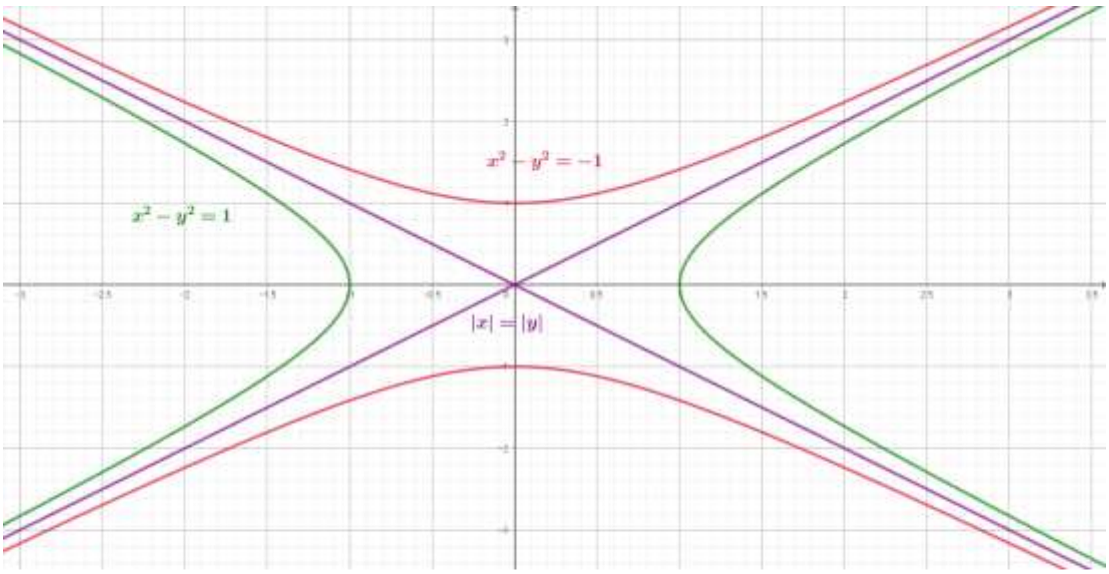
$$z = 0 \rightarrow \frac{x}{1+y^2} = 0 \rightarrow x = 0$$



d) $z = 0 \rightarrow x^2 - y^2 = 0 \rightarrow |x| = |y|$

$$z = 1 \rightarrow x^2 - y^2 = 1$$

$$z = -1 \rightarrow x^2 - y^2 = -1 \rightarrow y^2 - x^2 = 1 \ (es\ decir\ que\ está\ sobre\ el\ eje\ y)$$



e) $z = 1 \rightarrow \ln(x-y) = 1 \rightarrow y = x - e$

$$z = 0 \rightarrow \ln(x-y) = 0 \rightarrow y = x - 1$$

$$z = -1 \rightarrow \ln(x-y) = -1 \rightarrow y = x - e^{-1}$$